



MATERIAL SAFETY DATA SHEET OFDIURON 80% W/V SC

SECTION 1: CHEMICAL PRODUCT/COMPANY IDENTIFICATION

HIPLUS CO., LTD.

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Product name : Diuron 80% W/V SC

Common name : diuron

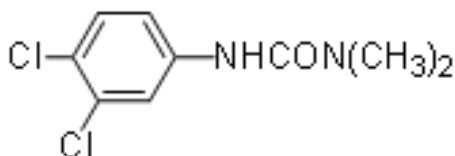
Chemical Name : (3-(3,4-dichlorophenyl)-1,1-dimethylurea (IUPAC)

CAS Number : 330-54-1

Molecular formula : C₉H₁₀Cl₂N₂O

Molecular Weight : 233.1

Structural formula :



Type of formulation : Suspension Concentrate (SC)

Use category : Herbicide

SECTION 2: COMPOSITION / INFORMATION ON INGREDIENTS

Composition	CAS no.	Content %
diuron	330-54-1	80
Inert Ingredients	- -	Balance



SECTION 3: HAZARDS IDENTIFICATION

Toxicity class:	EPA III (formulation), WHO III (active ingredient) A low toxicity herbicide.
Main hazard:	Toxic to fish and algae
Biological hazards:	Likely routes of exposure by skin contact, ingestion and inhalation. Eye contact: May cause eye irritation.
Skin contact:	Minimally toxic. Causes moderate irritation to skin.
Ingestion:	Minimally toxic. Methemoglobinemia may be noted in large ingestions.
Inhalation:	May cause irritation of nose and throat.

SECTION 4: FIRST AID MEASURES

Inhalation:

Remove source of contamination, or move victim to fresh air. Keep affected person warm and at rest. Treat symptomatically and supportively. If breathing is difficult, a qualified person should administer oxygen. Get medical attention immediately if effects persist.

Skin contact:

Move the victim to fresh air and remove all contaminated clothing, shoes and leather goods. Gently wipe off excess chemical. Wash affected skin areas gently and thoroughly with cold water and non-abrasive soap. Do not rub the skin. If irritation persists, obtain medical attention.

Eye contact:

Immediately flush the eyes with clean, gently flowing cold water for 20 minutes, holding the eyelid(s) open. If irritation persists, obtain medical attention.

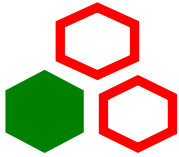
Ingestion:

Have victim rinse mouth thoroughly with water. Do not induce vomiting. Seek medical advice immediately, showing container and label.

Advice to physician:

There is no specific antidote. Treat symptomatically and supportively as and when required. Gastric lavage and catharsis could be considered when large quantities of the product has been ingested. Give oxygen if respiration is depressed.

Methemoglobinemia may be noted. If the patient is cyanotic and symptomatic, or methemoglobin level is greater than 30% in an asymptomatic patient, administer 1 to 2 mg/kg of 1% methylene blue slowly IV. Treat symptomatically and supportively as and when required.



SECTION 5: FIRE FIGHTING MEASURES

Fire and explosion hazard:

Flash point: None; water based. This material is not flammable.

Extinguishing agents:

Extinguish fires with carbon dioxide, dry powder, or alcohol-resistant foam. Water spray can be used for cooling of unaffected stock, but avoid water coming in contact with the product. Use as little water as possible. Use spray or fog. Solid stream may cause spreading.

Contain water used for fire fighting for later disposal.

Hazardous products of combustion:

In a fire, carbon oxides (CO, CO₂), nitrogen oxides (NOx) and corrosive fumes of chlorides may be formed.

Fire fighting:

Remove spectators from surrounding area. Remove container from fire area if possible. Contain fire control agents for later disposal. Use a recommended extinguishing agent for the type of surrounding fire. Avoid inhaling hazardous vapours and fumes from burning materials.

Keep upwind.

Personal protective equipment:

Fire may produce irritating or poisonous vapours (toxic fumes of carbon oxides, nitrogen oxides and corrosive fumes of chloride), mists or other products of combustion. Fire fighters and others that may be exposed should wear full protective clothing and self-contained breathing apparatus.

SECTION 6: HANDLING AND STORAGE

HANDLING:

Avoid inhalation of dust and contact with skin and eyes. Use with adequate ventilation. Wash hands before eating, drinking, chewing gum, smoking or using the toilet. Remove clothing immediately if the herbicide gets inside. Then wash skin thoroughly using a non-abrasive soap and put on clean clothing. Do not apply directly to areas where surface water is present, or to intertidal areas below the mean high water mark. Water used to clean equipment must be disposed of correctly to avoid contamination.

STORAGE:

Stable for at least 2 years when stored under dry dark warehouse conditions. Store in its original labeled containers in shaded, wellventilated area, away from heat, sparks and other sources of ignition. Not to be stored next to foodstuffs and water supplies. Keep out of reach of children and animals. Local regulations should be complied with.



SECTION 7: ACCIDENTAL RELEASE MEASURES

Personal precautions:

Avoid contact with skin and eyes. Do not breathe in spray or fumes. For personal protection see Section 8.

Environmental precautions:

Do not allow entering drains or watercourses. Spillage or uncontrolled discharges into water courses (or public waters) to be reported immediately to the Police and to the Department of Water/Environmental Affairs.

Occupational spill:

For small dry spills, sweep up with damp earth or sand or other suitable absorbent, such as sawdust, taking care not to raise a dust cloud. Place the material into a clean, dry container and cover for subsequent disposal. For larger spills contain spilled material for later disposal. In situations where product comes in contact with water, contain contaminated water for later disposal. Do not flush spilled material into drains. Keep spectators away and upwind.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

It is essential to provide adequate ventilation. The measures appropriate for a particular work site depend on how this material is used and on the extent of exposure. Ensure that control systems are properly designed and maintained. Comply with occupational safety, environmental, fire and other applicable regulations.

PERSONAL PROTECTIVE EQUIPMENT:

If engineering controls and work practices are not effective in controlling exposure to this material, then wear suitable personal protective equipment including approved respiratory protection.

Respirator:

It is usually safe to use the product without respirator. Where exposure through inhalation and eye contact may occur when handling and when preparing the spray mixture, wear a face mask. If the product is used in dusty or confined conditions or spillage and fire conditions a respirator suitable and approved for protection from dusts and mists of pesticides is adequate. Limitations of respirator use specified by the approved agency and the manufacturer must be observed.

Clothing:

Employee must wear appropriate protective (impervious) clothing and equipment to prevent repeated or prolonged skin contact with this substance.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Appearance: White viscous liquid

Odour: None

Explosive properties: Not explosive.

Flammability: Not flammable.



Flash point:	None.
Oxidising properties:	None.
pH:	4 – 7 (1% aqueous solution)
Bulk Density:	0.69.
Wettability:	10 sec max
Suspensibility:	60% min
Persistent foam:	15 ml max

SECTION 10: STABILITY AND REACTIVITY

Stability:

Stable under normal temperatures and storage conditions. Stable in neutral media. Hydrolyzed by acids and alkalis and at elevated temperatures.

Storage stability:

Stable for 2 years under normal warehouse conditions. Store at temperatures below 50 °C and above 0 °C.

Incompatibility:

Product is normally compatible with most herbicides, provided the mixing instructions are strictly adhered to. The quality of the water however, can affect the compatibility. Therefore, a compatibility test should be carried out with the water and the products to be used prior to mixing. Do not physically mix concentrate directly with other herbicides or pesticide concentrates; always dilute first. Avoid strong alkalis and acids.

Thermal decomposition:

Toxic oxides of carbon, nitrogen and corrosive fumes of chloride are released when the product decomposes on heating. Decomposes at 180 to 190 °C.

SECTION 11: TOXICOLOGICAL INFORMATION

Acute oral LD50	: > 2 000 mg/kg in rats.
Acute dermal LD50	: > 2 000 mg/kg in rats.
Acute inhalation LC50	: > 5 mg/l over 4 hours in rats.
Acute skin irritation	: The product is considered to be non-irritating to skin (rabbit).
Acute eye irritation	: The product may be moderately irritating to eyes (rabbit).
Dermal sensitization	: Product has a weak sensitisation potential (guinea pig).
Carcinogenicity	: Animal studies did not detect any carcinogenic activity.
Teratogenicity	: Animal studies did not detect any teratogenic activity.
Mutagenicity	: Animal studies did not detect any mutagenic activity.



SECTION 12: ECOLOGICAL INFORMATION

Degradability:

Diuron is moderately to highly persistent in soils. Residue half-lives are from 1 month to 1 year. Some pineapple fields contained residues 3 years after the last application. Mobility in the soil is related to organic matter and to the type of the residue. The metabolites are less mobile than the parent compound.

Diuron is relatively stable in neutral water. Microbes are the primary agents in the degradation of diuron in aquatic environments.

Diuron is readily absorbed through the root system of plants and less readily through the leaves and stems.

ECOTOXICOLOGY:

Birds : Practically non-toxic to birds.

LC50 (8 day diet) : 1730 mg/kg (bobwhite quail).
> 5000 mg/kg (Japanese quail).
> 5000 mg/kg (mallard ducklings).
> 5000 mg/kg (pheasant chicks).

Fish : Toxic to fish.

LC50 (96 hours) : Rainbow trout: 5.6 mg/l
Bluegill sunfish: 5.9 mg/l
Guppies: 25 mg/l

Daphnia: LC50 (48 hours): 12 mg/l

Bees : Not toxic.

This product is unlikely to bioaccumulate.

SECTION 13: DISPOSAL CONSIDERATIONS

Pesticide disposal:

Contaminated absorbents, used containers, surplus product, etc., should be buried in approved landfill. Hydrolysis under alkaline conditions is a suitable method to dispose of small quantities of the product. Never pour untreated waste or surplus products into public sewers or where there is any danger of run-off or seepage into water systems. Comply with local legislation applying to waste disposal.

Container disposal:

Emptied containers retain vapour and product residues and should be triple rinsed with water. Observe all labelled safeguards until container is cleaned, reconditioned or destroyed. Combustible containers should be disposed of in incinerators. Containers that are not to be reused should be punctured and transported to a scrap metal facility for recycling or disposal.



SECTION 14: TRANSPORTATION INFORMATION

UN NUMBER 3082

ADR/RID:

Shipping name: Environmentally hazardous substance, solid, n.o.s. (Diuron 800 g/l)

Substance ID no.: 3082

Hazard ID no.: 90

Label: 9

Item no.: 120(c)

IMDG/IMO

Proper shipping name: Environmentally hazardous substance, solid, n.o.s. (Diuron 800 g/l)

Class: 9

Packaging group: III

AIR/IATA

Proper shipping name: Environmentally hazardous substance, solid, n.o.s. (Diuron 800 g/l)

Class: 9

Packaging group: III

Y911 (30Kg)

911 No limit

UK classification Not Available

Tremcard number 90GM7-III

SECTION 15: REGULATORY INFORMATION

Symbol: N, Xn

Indication: Environmentally dangerous substance, Harmful.

Risk phrases:

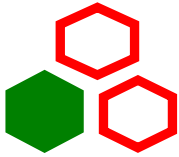
R 20/22 Harmful by inhalation and if swallowed.

R 36/37/38 Irritating to eyes, respiratory system and skin.

R 40 Limited evidence of a carcinogenic effect.

Carc. Cat. 3 Substance may cause concern for man owing to possible carcinogenic effects but in respect of which the available information is not adequate for making a satisfactory assessment.

R 50 Very toxic to aquatic organisms.



Safety phrases:

- | | |
|---------|---|
| S 2 | Keep out of reach children. |
| S 13 | Keep away from food, drink and animal feeding stuff. |
| S 24/25 | Avoid contact with skin and eyes. |
| S 36/37 | Wear protective clothing and gloves. |
| S 60 | This material and its container must be disposed of as hazardous waste. |
| S 61 | Avoid release to the environment. Refer to special instructions/safety data sheets. |

SECTION 16: OTHER INFORMATION

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